How To Use Electronics To Conserve Water

In recent times, several electronic devices, which are being developed to improve potable water availability, are becoming a part of the market and households. This article discusses different ways of using electronics to conserve, clean or produce water

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Water is one of the most precious ingredients for life, but with growing population, changing environmental conditions and changing lifestyles, potable water crisis is a reality all over the world. Only humans, among other living species that need water, are misusing potable water and its resources for such tasks as bathing and washing.

Clean and safe drinking water is a challenge for the growing global population and for millions of people around the world. To further complicate existing and increasing issues with supply, 85 per cent of diseases in the world are transmitted through non-potable water. The problem is represented below.

- About 2.1 billion people around the world live without access to safe drinking water.
- About 1.7 million children die annually due to lack of clean drinking water.
- About 443 million school days are lost annually due to water-related diseases.
- About 50 per cent of the world's population will live in water-stressed areas by 2025.

It is common knowledge that water destroys electronics and so the two do not go well together. But electronics has a number of applications for water, related to water conservation through water-level indicators/controllers, moisture sensing, waterless washing, water cleaning and purification, desalination and even generating water from air.

Growing scarcity of potable water all over the world, in general, and in many parts of India, in particular, needs the use of electronics—the present tool for everything—not only for water cleaning (waste water treatment or purification/desalination) and conservation, but to also make it fresh using renewable sources available on earth and in the atmosphere in meaningful quantities.

A solution is possible if a large quantity of potable water can be generated economically and in an environmental-friendly manner using electronics. So far, it has been a common belief that the global electronics industry has been slow to make large-scale innovative strides that address the use of electronics in dealing with potable water scarcity. However, in recent times, several electronic devices, which

Nearly waterless washing machine (right) that uses nylon polymer beads (below) for washing (Credit: www.forbes.com)